

YKP3089C017 Data Management Specification

Oct 2, 2026 · @Tracy Adu

Document control

This specification defines how data for the mock YKP3089C017 trial are collected, checked, queried and corrected without an EDC. Version 0.2 adds two amendments found while testing the checks on synthetic data.

Item	Detail
Study	YKP3089C017 (NCT01866111): Phase 2, double-blind, placebo-controlled trial of YKP3089 as adjunctive therapy for partial onset seizures
Protocol version	Amendment 3, 17-Jun-2019 (public protocol document)
Scope	Three forms from the double-blind phase: DM (Demographics & Enrollment), SZ (Seizure Diary), AE (Adverse Events). Open-label extension out of scope.
Data capture	No EDC. Data are held in spreadsheets; edit checks run offline in Python (run_edit_checks.py).
Status	Mock portfolio specification. Synthetic data only. Not affiliated with or endorsed by SK Life Science.

Version history

Version	Date	Change
0.2	02-Oct-2026	SZ06 anchored on the Visit 1 date. Baseline redefined as all diary days before Day 1. Rules added for never-randomized subjects. Data flow, query management and audit trail sections added.
0.1	29-Sep-2026	First draft: data dictionary, codelists, 19 edit checks, 2 review listings, protocol clarification log.

Study design and data conventions

Every date rule in this spec hangs on Study Day 1 = randomization (Visit 3). Visit 1 is Day -56, and there is no Day 0.

Visit	Target day	Window	Phase	Forms with new data
Visit 1	-56	±2 days	Screening, baseline starts	DM (consent, demographics); diary dispensed
Visit 2	-28	±2 days	Baseline	SZ, AE
Visit 3	1	Defines Day 1	Randomization	SZ, AE; DM randomization date and first-dose-at-site flag
Visit 4	15	±2 days	Titration	SZ, AE; DM first dose date
Visit 5	29	±2 days	Titration	SZ, AE
Visit 6	43	±2 days	End of titration	SZ, AE
Visit 7	71	±3 days	Maintenance	SZ, AE
Visit 8	99	±3 days	Maintenance	SZ, AE
Visit 9	127	±3 days	End of maintenance or early termination	SZ, AE
Visit 10	162	±3 days	Follow-up, 14 days after last dose	SZ, AE

Phone contacts on Days 8, 22 and 36 collect AEs but no diary data (protocol 9.4).

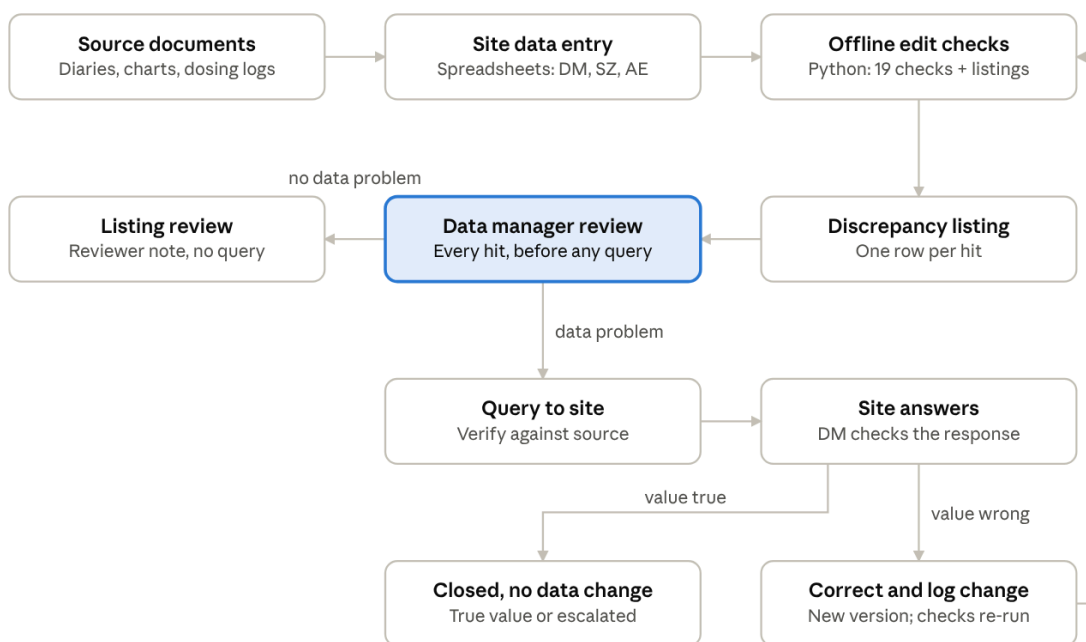
Conventions

- Dates are DD-MMM-YYYY (e.g., 05-MAR-2026).
- Variable names follow CDISC CDASH where a standard exists. The seizure diary has no CDASH standard, so its fields are sponsor-defined.
- SUBJID is the 8-digit screening number CCSSSNNN (country, site, order screened). IDs are stored as text so leading zeros survive.
- Treatment assignment never appears in data management datasets. Review is done blinded.

Data flow

Without an EDC, every export goes through the same offline checks, and a person decides what each hit needs before anything goes to a site.

Every hit is reviewed by a person, and every fix re-runs the checks



offline data flow · query loop and re-run

In the test run, 25 hits became 20 queries and 3 listing reviews. 21 corrections were logged, and the re-run on the corrected data found 0 new discrepancies.

Data dictionary

The three forms hold 42 fields: 13 in DM, 12 in SZ and 17 in AE. Derived fields are calculated, never entered by the site (rules in Derivations).

DM — Demographics & Enrollment (one record per subject)

Variable	eCRF question	Type and values	Required	Standard	Protocol
STUDYID	Study identifier	Text; pre-populated YKP3089C017	Auto	CDASH	Title page
SITEID	Site number	Text, 3 digits	Yes	CDASH	8.2.1.1

Variable	eCRF question	Type and values	Required	Standard	Protocol
SUBJID	Screening number	Text, 8 digits CCSSSNNN	Yes	CDASH	8.2.1.1
RFICDAT	Date informed consent signed	Date	Yes	Sponsor (full build: DS form)	7.3 #3; 9.1.1
BIRTHDAT	Date of birth	Date	Yes	CDASH	7.3 #1
AGE	Age at informed consent (years)	Integer	Derived	CDASH (derived)	7.3 #1
SEX	Sex	SEX codelist	Yes	CDASH	7.3 #1
CHILDPOT	Is the subject of childbearing potential?	NY codelist	If SEX = F; blank if M	Sponsor	7.3 #10; 7.5
ETHNIC	Ethnicity	ETHNIC codelist	Yes	CDASH	—
RACE	Race (check all that apply)	RACE codelist, multi-select	Yes	CDASH	—
RANDDAT	Date of randomization (Study Day 1)	Date	If randomized	Sponsor (from IWRS)	6.1.1; 9.1.3
EXSTDAT	Date of first dose of study drug	Date	If randomized	CDASH (EX domain)	6.1.1; 9.1.3
FDOSSITE	Was the first dose given at the site on Day 1?	NY codelist	If randomized	Sponsor	9.1.3

SZ — Seizure Diary transcription (one record per subject per diary day)

Variable	eCRF question	Type and values	Required	Standard	Protocol
SUBJID	Screening number	Text, 8 digits	Auto	CDASH	8.2.1.1

Variable	eCRF question	Type and values	Required	Standard	Protocol
VISIT	Visit at which diary was returned	VISIT codelist	Yes	CDASH	Table 1
VISDAT	Date diary was returned	Date	Yes	CDASH	Table 1
SZDAT	Diary date	Date; one record per calendar day	Yes	Sponsor	9.1.1
SZDIARY	Did the subject record diary data for this day?	NY codelist	Yes	Sponsor	6.2.1
SZSPM	Simple partial seizures WITH motor component	Integer 0–99	If SZDIARY = Y	Sponsor	6.2.1; 7.3 #6
SZSPNM	Simple partial seizures WITHOUT motor component	Integer 0–99	If SZDIARY = Y	Sponsor	6.2.1
SZCP	Complex partial seizures	Integer 0–99	If SZDIARY = Y	Sponsor	6.2.1
SZSG	Secondarily generalized seizures	Integer 0–99	If SZDIARY = Y	Sponsor	6.2.1
SZCOUNT	Countable seizures for the day	Integer	Derived	Derived	6.2.1
SZDY	Study day of diary date	Integer	Derived	Derived	Table 1
SZPHASE	Study phase	Text	Derived	Derived	Table 1 note 1

Blank is not zero. A day with no seizures is SZDIARY = Y with all counts = 0. A blank or missing diary page is SZDIARY = N with counts left blank, and that day drops out of the seizure-rate denominator.

AE — Adverse Events (one record per event)

Variable	eCRF question	Type and values	Required	Standard	Protocol
AESPID	AE number	Integer, sequential; never reused	Auto	CDASH	—
AETERM	Adverse event (verbatim term)	Text, 200 characters	Yes	CDASH	10.3; 10.4
AESTDAT	Start date	Date; UN allowed for unknown day or month	Yes	CDASH	10.3
AESTTIM	Start time	HH:MM (24-hour) or UNK	Yes	CDASH	10.3
AEONGO	Is the event ongoing?	NY codelist	Yes	CDASH	10.4
AEENDAT	End date	Date	If AEONGO = N	CDASH	10.3
AESEV	Severity	AESEV codelist	Yes	CDASH	10.3
AEREL	Relationship to study drug	AEREL codelist (protocol 5-point scale)	Yes	CDASH variable, sponsor codelist	10.3
AEACN	Action taken with study drug	ACN codelist	Yes	CDASH	6.1.2.1; 6.1.5
AESER	Is the event serious?	NY codelist	Yes	CDASH	10.1.2
AESDTH	Results in death	NY codelist	If AESER = Y	CDASH	10.1.2
AESLIFE	Is life-threatening	NY codelist	If AESER = Y	CDASH	10.1.2

Variable	eCRF question	Type and values	Required	Standard	Protocol
AESHOSP	Requires or prolongs hospitalization	NY codelist	If AESER = Y	CDASH	10.1.2; 10.2
AESDISAB	Persistent or significant disability/incapacity	NY codelist	If AESER = Y	CDASH	10.1.2
AESCONG	Congenital anomaly or birth defect	NY codelist	If AESER = Y	CDASH	10.1.2
AESMIE	Other important medical event	NY codelist	If AESER = Y	CDASH	10.1.2
AEOUT	Outcome	OUT codelist	Yes	CDASH	10.3

Seizures belong in the diary. They are AEs only when they show a measurable increase in frequency or duration, or a pattern unlike the subject's usual seizures (protocol 10.4.1). An emergency room visit alone is not hospitalization (10.2).

Codelists

Nine codelists cover every coded field. CDISC controlled terminology is used where it exists; AEREL and VISIT come from the protocol.

Codelist	Submitted values	Source
NY	Y, N	CDISC CT
SEX	M, F	CDISC CT (subset)
ETHNIC	HISPANIC OR LATINO; NOT HISPANIC OR LATINO; NOT REPORTED; UNKNOWN	CDISC CT
RACE	AMERICAN INDIAN OR ALASKA NATIVE; ASIAN; BLACK OR AFRICAN AMERICAN; NATIVE HAWAIIAN OR OTHER PACIFIC ISLANDER; WHITE; NOT REPORTED; UNKNOWN	CDISC CT
AESEV	MILD; MODERATE; SEVERE	CDISC CT, protocol 10.3 definitions

Codelist	Submitted values	Source
AEREL	UNRELATED; REMOTE; POSSIBLE; PROBABLE; DEFINITE	Sponsor (protocol 10.3)
ACN	DOSE NOT CHANGED; DOSE REDUCED; DRUG INTERRUPTED; DRUG WITHDRAWN; NOT APPLICABLE; UNKNOWN	CDISC CT (subset)
OUT	RECOVERED/RESOLVED; RECOVERING/RESOLVING; NOT RECOVERED/NOT RESOLVED; RECOVERED/RESOLVED WITH SEQUELAE; FATAL; UNKNOWN	CDISC CT
VISIT	VISIT 1 to VISIT 10; UNSCHEDULED	Protocol Table 1

Severity uses the protocol's definitions: mild is easily tolerated, moderate interferes with daily activity, severe is incapacitating and needs medical intervention. Events that start before dosing are UNRELATED.

Derivations

Derived values are recalculated whenever an input changes, and each recalculation is logged in the audit trail as a system change.

Variable or window	Rule	Notes
AGE	Whole years between BRTHDAT and RFICDAT	Inclusion 1 requires 18–70
SZCOUNT	SZSPM + SZCP + SZSG	Blank when SZDIARY = N or any of the three counts is blank. SZSPNM is not counted (protocol 6.2.1).
SZDY	SZDAT – RANDDAT + 1 when SZDAT ≥ RANDDAT; otherwise SZDAT – RANDDAT	No Day 0. Blank for never-randomized subjects.
SZPHASE	BASELINE: every diary day before Day 1. DAY 1. TITRATION: Day 2–43. MAINTENANCE: Day 44–127.	v0.2: baseline is no longer fixed at Day –56, because a delayed Visit 3 makes it longer. Never-randomized subjects are BASELINE throughout.

Variable or window	Rule	Notes
	POST-TREATMENT: Day 128 or later.	
Day 1 seizures	Excluded from efficacy counts, unless FDOSSITE = Y; then seizures after the first dose count	Table 1 note 1; 9.1.3
Baseline window (SZ06)	Visit 1 date (RFICDAT) through the day before randomization, or before Visit 3 if not randomized	v0.2: anchored on Visit 1, because the check runs before RANDDAT exists. Weeks 1–4 = first 28 days; weeks 5–8 = the rest.
Seizure-free run (SZ06)	Longest run of consecutive recorded days with SZCOUNT = 0	Missing days (SZCOUNT blank) break a run; they don't extend it.

The primary endpoint shows why a blank must never become a zero. Each period's rate divides seizures by days with usable diary data (protocol 6.2.1):

$$\text{Rate per 28 days} = \frac{\sum \text{SZCOUNT}}{\{\text{days with SZDIARY} = Y\}} \times 28$$

$$\text{Percent change} = 100 \times \frac{D - B}{B}$$

B is the baseline rate and D is the double-blind rate. A blank day entered as zero adds a false seizure-free day to the denominator. That lowers the rate for whichever period it falls in, which biases the percent change.

Edit checks

Nineteen programmed checks run against every data export: 9 hard, 8 soft and 2 with a hard part and a soft part. Without an EDC, a hard check can't block entry, so it marks a value that can never be right and the query asks for a correction. A soft check marks a value that might be true and asks the site to verify it.

ID	Rule (fires when...)	Hard / Soft	Query text (site-facing)	Protocol
DM01	AGE < 18 or > 70	Soft	Age at consent ([AGE]) is outside the protocol range of 18–70 years (Inclusion 1). Please verify date of birth and date of consent against source.	7.3 #1
DM02	SUBJID is not 8 digits, or digits 3–5 ≠ SITEID	Hard	Screening number must be 8 digits, with digits 3–5 matching the site	8.2.1.1

ID	Rule (fires when...)	Hard / Soft	Query text (site-facing)	Protocol
			number. Please correct.	
DM03	Female with CHILDPOT blank, or male with CHILDPOT entered	Hard	Childbearing potential must be answered for female subjects and left blank for male subjects. Please review.	7.3 #10
DM04	BRTHDAT $\geq$ RFICDAT, or RFICDAT in the future	Hard	Date of birth is on or after the consent date, or the consent date is in the future. Please verify against source.	—
DM05	RANDDAT $\leq$ RFICDAT (hard); gap < 54 or > 58 days (soft)	Hard + Soft	Randomization date is [N] days after informed consent. The protocol specifies an 8-week baseline (Day -56 to Day 1, ± 2 days). Please verify both dates.	6.1; Table 1
DM06	EXSTDAT $\neq$ RANDDAT + 1 (or $\neq$ RANDDAT when FDOSSITE = Y)	Soft	Date of first dose is [N] day(s) after randomization. Per protocol, dosing begins the morning after randomization, or on Day 1 if given at the site. Please verify.	6.1.1; 9.1.3
SZ01	SZDIARY = Y with any count blank, or SZDIARY = N	Hard	Diary data recorded for this day is [Y/N], but the seizure counts are	6.2.1

ID	Rule (fires when...)	Hard / Soft	Query text (site-facing)	Protocol
	with any count entered		[missing/entered]. Please review against the diary page.	
SZ02	Same SUBJID and SZDAT on more than one record	Hard	A diary record already exists for this date. Please remove the duplicate or correct the date.	—
SZ03	SZDAT before RFICDAT or after VISDAT	Hard	Diary date is before informed consent or after the visit at which the diary was returned. Please verify the date against the diary.	9.1.1
SZ04	Any single count > 20 in one day	Soft	[N] seizures ([type]) are recorded on [date]. Please confirm against the source diary.	7.4 #4
SZ05	Any calendar day from Visit 1 to the latest diary return has no record	Soft	No diary record exists for [dates]. Please enter the diary data, or enter records with 'diary data recorded' = No if the diary was not completed.	6.2.1
SZ06	Baseline: < 8 countable seizures, < 3 in weeks 1–4 or 5–8, or a seizure-free run > 25 days (v0.2 window)	Soft	Per the diary data entered, the subject does not appear to meet the baseline seizure criteria (Inclusion 6): [detail]. Please verify the diary	7.3 #6

ID	Rule (fires when...)	Hard / Soft	Query text (site-facing)	Protocol
			entries against source.	
AE01	AEENDAT before AESTDAT	Hard	AE end date is before the start date. Please verify both dates against source.	10.3
AE02	Ongoing with an end date, or resolved without one (hard); outcome disagrees with ongoing status (soft)	Hard + Soft	The ongoing status, end date and outcome for this event are inconsistent. Please review.	10.3; 10.4
AE03	AESER = Y with no criterion = Y, or AESER = N with any criterion = Y	Hard	This event's seriousness and seriousness criteria are inconsistent. Please review.	10.1.2
AE04	AESTDAT before RFICDAT	Soft	AE start date is before informed consent. Please verify the start date, or confirm whether this condition should be recorded as medical history.	10.1.4; 10.1.5
AE05	AESTDAT before EXSTDAT and AEREL ≠ UNRELATED	Soft	This event started before the first dose of study drug, and relationship is recorded as [AEREL]. Please review the relationship assessment per	10.3

ID	Rule (fires when...)	Hard / Soft	Query text (site-facing)	Protocol
			protocol section 10.3.	
AE06	AETERM contains seizure, convulsion or status epilepticus	Soft	Seizures are normally recorded in the seizure diary. Please confirm this event meets the AE definition in protocol section 10.4.1.	10.4.1
AE07	AEOUT = FATAL without AESER = Y and AESDTH = Y	Hard	Outcome is Fatal, but the event is not marked serious with 'results in death'. Please review.	10.1.2

The SZ04 threshold of 20 and the DM05 window are working assumptions to confirm with the medical monitor.

Review listings

Some hits need a person's judgment rather than an automatic query. These go to a reviewer first, and a query goes out only if the review finds a real data problem.

ID	What it lists	Runs	Reviewer action
ML01	AEs with AEACN = DOSE REDUCED that started in Week 1 (Days 2–8) or after Week 8 (Day 58 or later)	Every 2 weeks	Confirm the reduction date in dosing records. The AE start date isn't the reduction date, so this can't be a programmed check. Reductions in Week 1 or after Week 8 are deviations.
ML02	Every AE with AESER = Y	Monthly, and before database lock	Reconcile against the safety database: term, onset date, seriousness criteria, relationship, outcome. Query

ID	What it lists	Runs	Reviewer action
			whichever source is wrong.
SZ06 (not randomized)	Screen failures whose baseline fails Inclusion 6	Before each randomization	Confirm the subject wasn't randomized. A failing baseline in a screen failure is the expected result, so no query is needed.

When checks are re-run, hits already closed as confirmed correct, escalated, or reviewed in a listing are not queried again. Only new hits go to review.

Query management

A person reviews every hit before a query is sent, and a query closes only after data management has checked the answer.

1. **Review first.** Each hit is classed as a query, a listing review, or an expected result (for example, a screen failure).
2. **One root cause, one query.** When a single error fires two checks, the site gets one query that names both. In testing, a mistyped diary month fired SZ03 and SZ05 and was raised once.
3. **Neutral wording.** A query states what was found and the rule it breaks, and asks the site to verify against source. It never suggests the answer.
4. **Complete responses.** The site says which source document was checked and what was changed, or why the value is correct as entered.
5. **Re-query when unresolved.** If the answer doesn't settle the discrepancy, the query is re-issued with a clearer question. Example: a site answered "no seizures that day" for a blank diary page, which needs counts left blank rather than zeros.
6. **Close after verification.** Data management confirms the correction in the data, or accepts a documented confirmation, before closing.
7. **Escalate deviations.** When confirmed-correct data reveal a protocol problem (an eligibility failure or an out-of-window visit), the query closes and the issue goes to the CRA and medical monitor as a deviation.

Queries are numbered QRY-001 onward. Status moves Open → Answered → Closed, and a re-query sends it back to Open. The proposed turnaround target is a site response within 7 days. The metrics report percent answered on time by site.

Audit trail and data changes

No value changes without a logged reason, and the old value is never lost.

Audit field	Content
Audit ID	AT-001 onward

Audit field	Content
Date of change	Date the site answered the query and made the change
SUBJID, form, record, field	Exactly what changed
Old value and new value	Both kept. A deletion records the old record and "(deleted)"; an added record records "(no record)".
Changed by	Role, not name (e.g., Site 002 coordinator). Derived recalculations are "System (derived)".
Reason	The query ID and the site's explanation

- Deleted records stay in the audit trail. AESPID numbers are never reused, so gaps are expected.
- Each corrected dataset is a new version (v1.0 before review, v1.1 after). Earlier versions are kept.
- The log's content follows the principles of 21 CFR Part 11: changes are attributable, dated, and keep the prior value. A manually kept spreadsheet log is not itself a Part 11-compliant system; a validated EDC generates this trail automatically.

Protocol clarification log

Building the spec surfaced five places where the protocol contradicts itself or is silent. Each has a working assumption until the sponsor answers.

ID	Section	Issue	Assumption used	Status
Q1	10.1.4	The AE reporting period is given as 21 days and, in the next sentence, as 30 days after the last dose.	30 days after last dose	Open
Q2	6.2.1 vs 7.3 #6	The endpoint excludes simple partial seizures "without a motor/visual component", implying visual ones count. Inclusion 6 counts only those "with motor component".	Count motor only; collect non-motor separately (SZSPNM) so counts can be re-derived	Open
Q3	7.3 #7c	Points to Exclusion 12 for rescue	Exclusion 13 is intended	Open

ID	Section	Issue	Assumption used	Status
		benzodiazepines; the rescue criterion is Exclusion 13.		
Q4	10.4.1 vs Table 7	SAE reporting is "within one business day" in the text but "within 24 hours" in Table 7.	24 hours (the stricter)	Open
Q5	7.4 #4	Seizure clusters are defined only as an exclusion. There is no rule for counting clusters on-study.	No cluster field; SZ04 prompts review of very high counts	Open
Q6	Table 1 note I; 9.1.3	Day 1 seizures are not counted, except those after a first dose given at the site.	FDOSSITE field added	Handled in spec

Related files

File	What it holds
YKP3089C017_DM_Spec_v0.1.xlsx	Working spec tables (v0.1). This document supersedes it where they differ.
run_edit_checks.py	Python implementation of every check and listing in this spec, including the v0.2 amendments
YKP3089C017_Synthetic_Data_v1.xlsx	Synthetic test data before review: 20 subjects, 3,930 diary records, 34 AEs
YKP3089C017_Planted_Issues_KEY.xlsx	Answer key for the 21 errors planted to test the checks
YKP3089C017_Data_Review_Package.xlsx	Discrepancy listing, query log, listing review, audit trail, re-run results and metrics
YKP3089C017_Synthetic_Data_v1.1_post_review.xlsx	Data after all query resolutions were applied

Source protocol: YKP3089C017 Clinical Study Protocol, Amendment 3, 17-Jun-2019 (public document).